Uncertainty Analysis

How do we record measurements in experiments and how do we process them afterwards? How to deal with uncertainty in measurements and calculations? There are a few rules.

Example: Let's say we have done measurements on two quantities and the results are

$$a = 3.00 \pm 0.03 \text{ (m)} \qquad b = 1.00 \pm 0.02 \text{ (m)}$$

Uncertainty within 3 cm. Uncertainty within 2 cm.

$$\Delta a = 0.03 \text{ (m)} \qquad \Delta b = 0.02 \text{ (m)}$$

1) How much is the uncertainty in $S = a + b$ and $S = a - b$?

For addition and subtraction, add uncertainties.

$$S = a + b = 4.00 \text{ (m)}$$

$$\Delta S = \Delta a + \Delta b = 0.05 \text{ (m)}$$

$$S = 4.00 \pm 0.05 \text{ (m)}$$

$$S = a - b = 2.00 \text{ (m)}$$

$$\Delta S = \Delta a + \Delta b = 0.05 \text{ (m)}$$

$$S = 2.00 \pm 0.05 \text{ (m)}$$

Uncertainty Analysis

Example: Let's say we have done measurements on two quantities and the results are

$a = 3.00 \pm 0.03$ (m) $\qquad b = 1.00 \pm 0.02$ (m)

$\Delta a = 0.03$ (m) $\qquad \Delta b = 0.02$ (m)

2) How much is the uncertainty in $S = 2a - 3b$?

$S = 2a - 3b = 3.00$ (m) $\qquad \Delta S = 2\Delta a + 3\Delta b = 0.12$ (m) $\qquad S = 3.00 \pm 0.12$ (m)

3) How much is the uncertainty in $S = ab$?

For Multiplication and division, add percentage uncertainties.

$$\%a = \frac{\Delta a}{a} \times 100 = 1\ \%$$

$$\%b = \frac{\Delta b}{b} \times 100 = 2\ \%$$

$$\%S = \%a + \%b = 3\ \%$$

$S = 3.00$ (m^2)

$\Delta S = \%S \cdot S = 0.09$ (m^2)

$S = 3.00 \pm 0.09$ (m^2)

Uncertainty Analysis

Example: Let's say we have done measurements on two quantities and the results are

$a = 3.00 \pm 0.03$ (m) $\qquad b = 1.00 \pm 0.02$ (m)

$\Delta a = 0.03$ (m) $\qquad \Delta b = 0.02$ (m)

4) How much is the uncertainty in $S = a^2 b^{-3}$?

$$\%a = \frac{\Delta a}{a} \times 100 = 1\ \%$$

$$\%b = \frac{\Delta b}{b} \times 100 = 2\ \%$$

$$\%S = 2\%a + 3\%b = 8\ \%$$

$S = 9.00$ (m^{-1})

$\Delta S = \%S \cdot S = 0.72$ (m^{-1})

$S = 9.00 \pm 0.72$ (m^{-1})

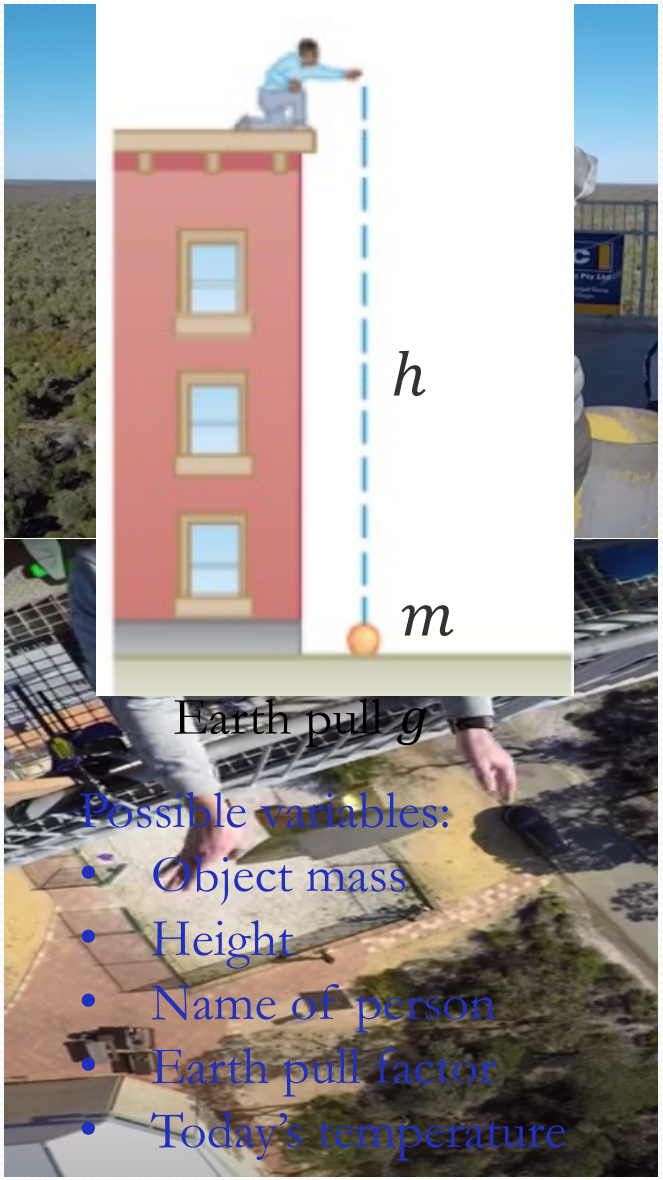
5) How much is the uncertainty in $S = 5a^2 b^{-3}$?

$$\%S = 2\%a + 3\%b = 8\ \%$$

$S = (5)(9.00) = 45$ (m^{-1})

$S = 45.00 \pm 3.60$ (m^{-1})

Dimensional Analysis



A powerful procedure to obtain physical laws of new phenomena using basic principles and intuition. Many important laws were established this way!

Let's assume that we do not know the equations of motion. We want to find them out ourselves!

Free-fall experiment

We need a law for the time (t) required for any object of mass (m) and height (h) above the ground, to reach the ground.

$$t \propto h^x m^y g^z$$

$$T^1 \propto L^x M^y \left(\frac{L}{T^2}\right)^z \Rightarrow \begin{aligned} x + z &= 0 \\ y &= 0 \\ -2z &= 1 \end{aligned}$$

$$z = -1/2$$

$$x = 1/2$$

$$t = C\sqrt{\frac{h}{g}}$$

Chapter 2: Motion in a Straight Line
(Constant Acceleration)

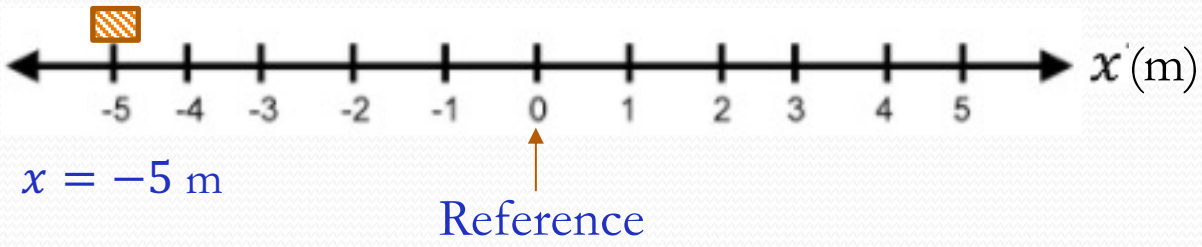
The following subjects will be discussed

- Definitions.
- Equations of Motion.
- A Special Case: Free Fall.
- Examples.

Definitions

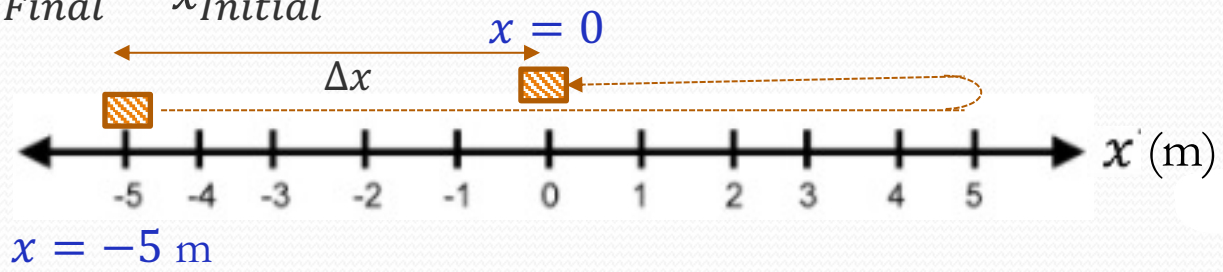
We start with defining the motion variables and parameters

1) Position: The location of an object with respect to a reference. In a coordinate system, the reference is taken as the origin.



2) Displacement: The change in the object position. (Final position-initial position)

$$\Delta x = x_{Final} - x_{Initial}$$



$$\Delta x = 0 - (-5) = 5\ m$$

Positive/negative Displacement

Sense of Direction

Distance?

Definitions

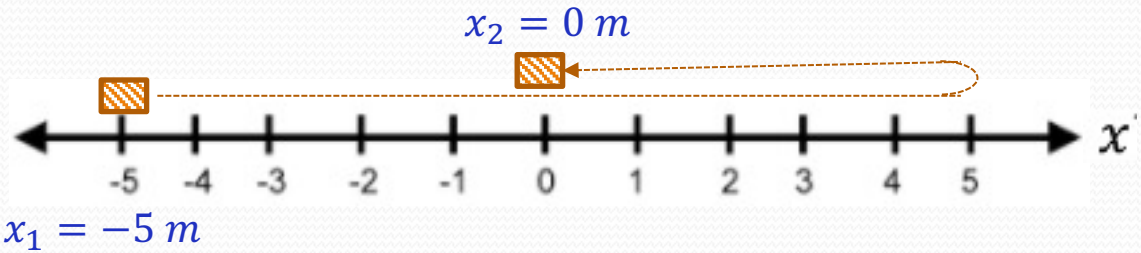
3) Average Speed: The total distance travelled by total time.

$$Speed_{ave} = \frac{total\ distance}{total\ time} \quad \text{(m/s)}$$

Always positive
No sense of direction

4) Average Velocity: The net displacement done by time interval.

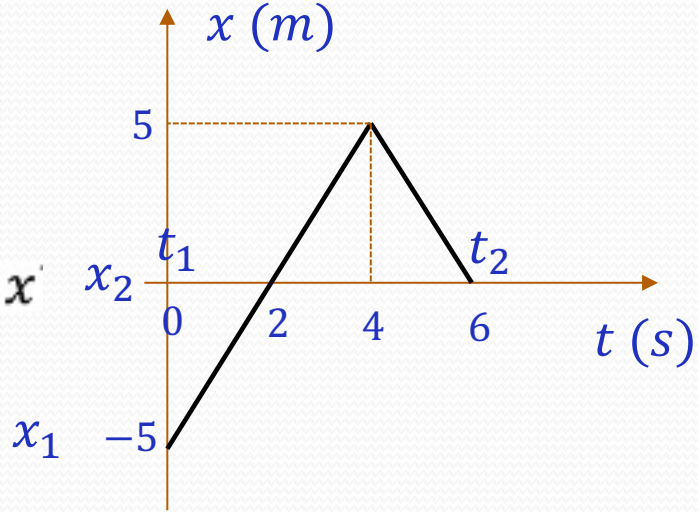
$$v_{ave} = \frac{\Delta x}{\Delta t} \quad \text{(m/s)}$$



$t_1 = 0$ $\qquad$ $t_2 = 6\ s$

$$v_{ave} = \frac{0-(-5)}{6-0} = \frac{5}{6}\ \text{m/s}$$

$$Speed_{ave} = \frac{15}{6-0} = \frac{15}{6}\ \text{m/s}$$

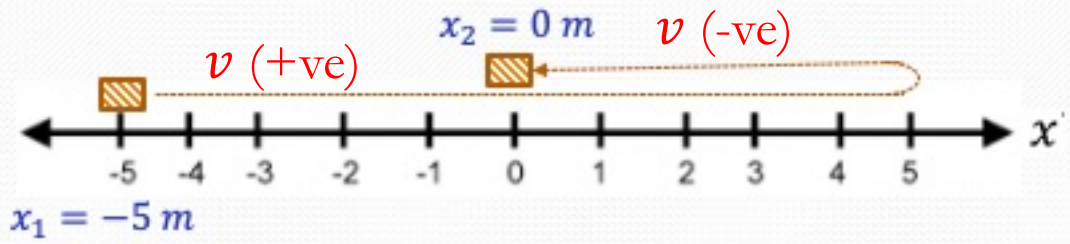


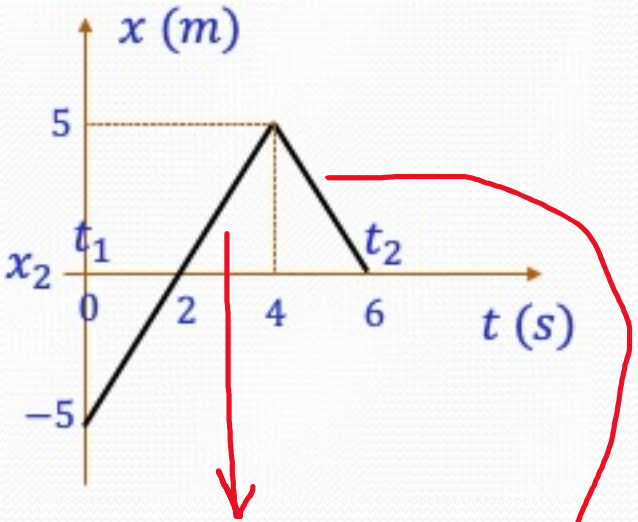
v_{ave}(+ve): motion (on average) to the right
v_{ave}(-ve): motion (on average) to the left

Definitions

5) Instantaneous Velocity: The rate of change of the position function at a given time.
The slope of the position-time curve at a point in time.

$$v = \frac{dx}{dt} \qquad \text{(m/s)}$$





v (+ve): motion at that specific time is to the right.
v (-ve): motion at that specific time is to the left.

$$v = \frac{5-(-5)}{4-0} = \frac{10}{4}\ \text{m/s}$$

$$v = \frac{0-5}{6-4} = \frac{-5}{2}\ \text{m/s}$$

Definitions

6) Average Acceleration: The net change in instantaneous velocity over time interval.

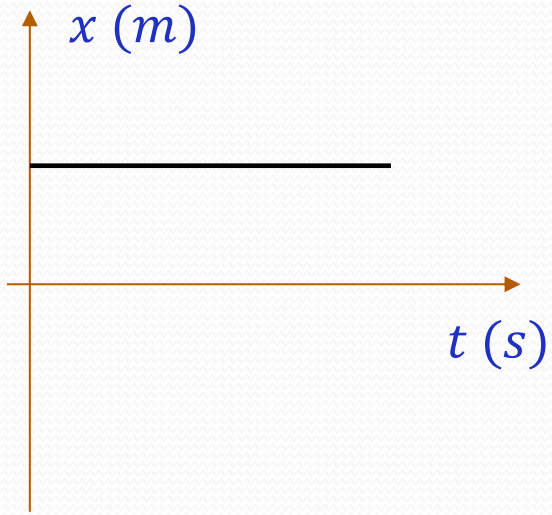
$$a_{ave} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1} \qquad (m/s^2)$$

7) Instantaneous Acceleration: The rate of change of the velocity at a given time.
The slope of the velocity-time curve at a point in time.

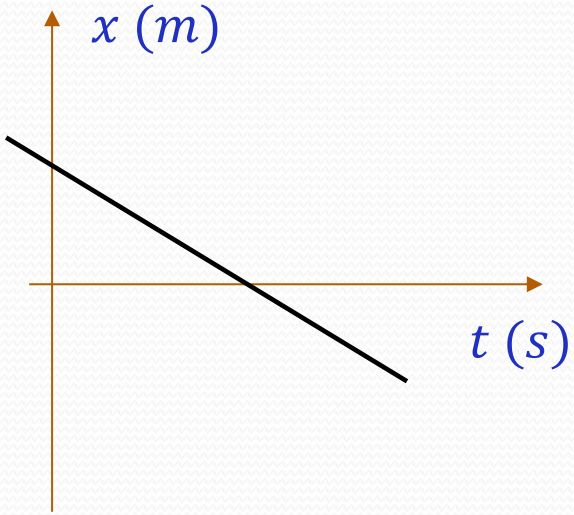
$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} \qquad (m/s^2)$$

a (+ve): speed at that specific time is increasing.
a (-ve): speed at that specific time is decreasing.

Remarks

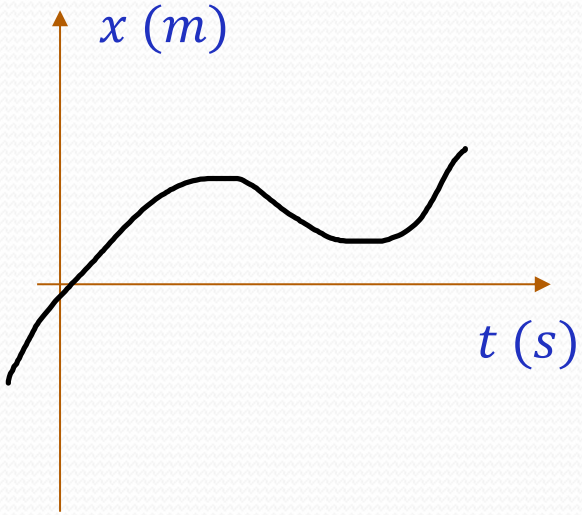


Instantaneous velocity is zero.
Object not moving.



Instantaneous velocity is constant (negative).
Object moving with zero acceleration.

Object moving to the left.



Instantaneous velocity is varying.
Object moving with some acceleration.

Q & A

Send me an email if
you have any
questions.